INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY 9729/02

Paper 2 Structured Questions **27 August 2018**

Candidates answer on the question paper. **2 hours**

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group on all the work you hand in.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

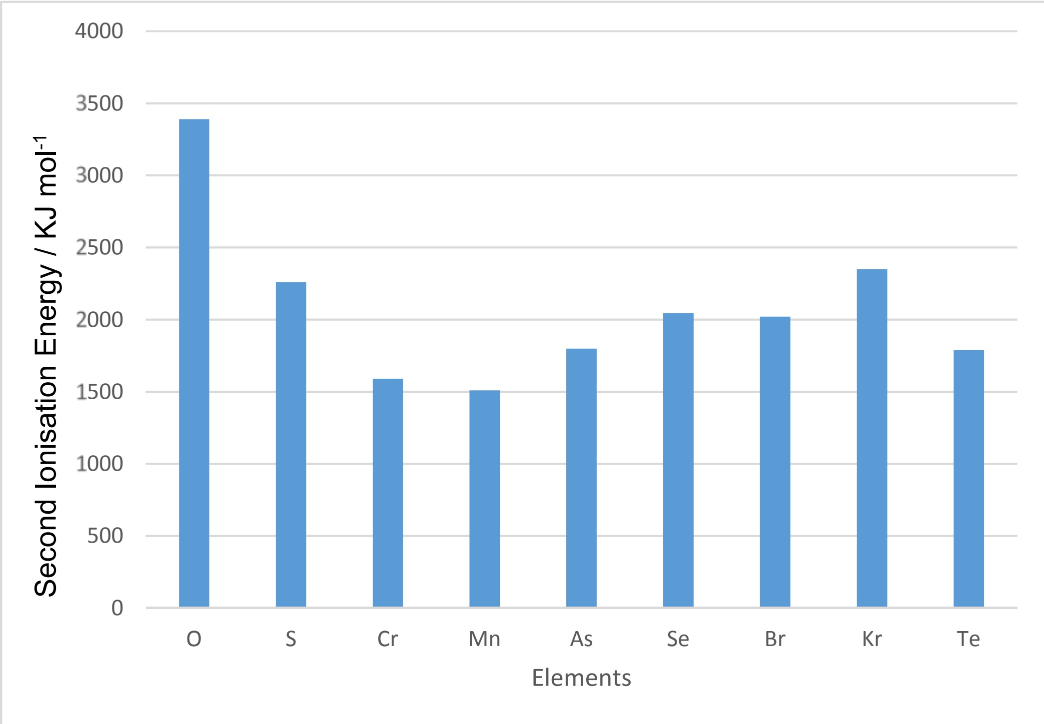
The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	9
2	8
3	6
4	15
5	13
6	12
7	12
Significant Figures and Units	
Handwriting and Presentation	
Total	75

This document consists of **19** printed pages **1** blank page.

1 **(a)** The following diagram shows the second ionisation energy of some elements.



(i) Explain why the second ionisation energies of elements O, S, Se and Te show a decreasing trend.

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....[1]

(ii) Explain why the second ionisation energy of Br is lower than that of Se.

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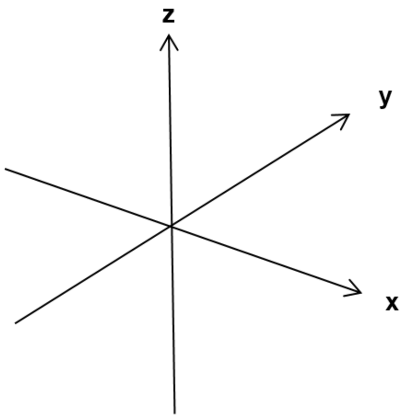
....[1]

(iii) Write the full ground state electronic configuration for Cr.

....[1]

(iv) Draw the d_{xy} orbital in the axes provided below. [1]



(b) Oxygen–oxygen bond lengths in some molecules are given below:

Molecule	Bond Length
Oxygen, O_2	0.121 nm
Hydrogen peroxide, H_2O_2	0.149 nm
Ozone, O_3	0.128 nm

(i) Draw the structure of the molecule ozone, O_3. [1]

(ii) With reference to the data given above, comment and explain the oxygen–oxygen bond length in ozone as compared to oxygen and hydrogen peroxide.

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(iii) Explain clearly why hydrogen peroxide is a liquid while oxygen is a gas at room conditions in terms of structure and bonding.

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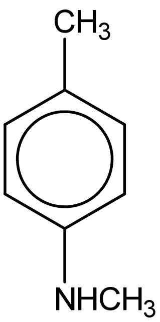
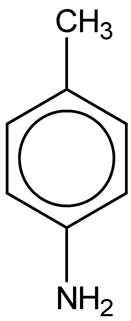
[Total: 9]

2 Compounds containing nitrogen are important to life and have applications in science and medicine.

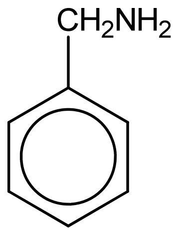
Three nitrogeneous bases have the following structures.



N,4-dimethylbenzenamine



4-methylphenylamine



benzylamine

(a) **(i)** Arrange the three bases above in increasing order of pK_b.

…………………………………………………………………………………………[1]

(ii) Explain in terms of their molecular structures why benzylamine and 4-methylphenylamine have different pK_b values.

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…………………………………………………………………………………………[2]

(iii) Outline how *N*,4-dimethylbenzenamine may be produced from 4-methylphenylamine.

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……………………………………………………………………………………………...[1]

(b) The pK_b of benzylamine is 4.66.

25.0 cm^3 of 0.025 mol dm^{-3} benzylamine was completely neutralised by dilute hydrochloric acid of the same concentration. The salt formed reacts with water and the pH of the resultant solution is less than 7.

(i) Write the equation to show the reaction between the salt formed and water.

……………………………………………………………………………………………...[1]

(ii) With reference to your equation in **(b)(i)**, write an expression for the acid dissociation constant of the salt.

[1]

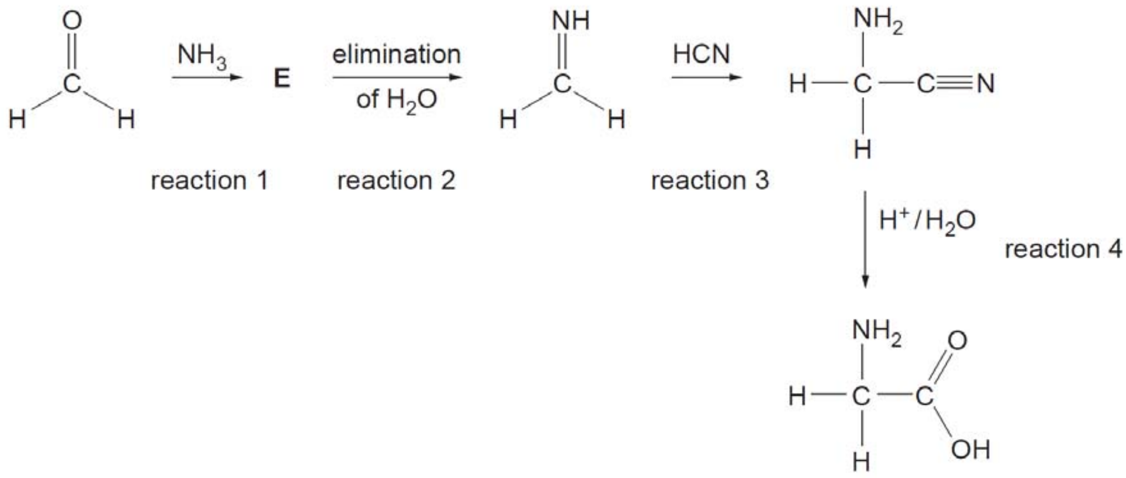
(iii) Hence, determine the pH of the resultant solution.

[2]

[Total: 8]

3 Methanal is a colorless, strong-smelling gas used in making building materials and many household products.

The Strecker synthesis is a route to preparing amino acids. Glycine, 2-aminoethanoic acid, can be prepared from methanal in this way. This is shown in the four steps reaction scheme below.



2-aminoethanoic acid

(a) **(i)** Suggest the role of ammonia in this synthesis.

…………………………………………………………………………………...[1]

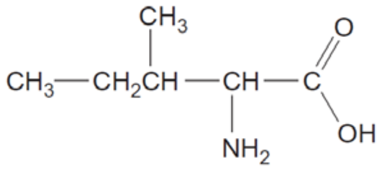
(ii) Compound **E** has the molecular formula CH_5NO.
Suggest a structure for compound **E**.

[1]

(iii) State the type of reaction for reaction 4.

…………………………………………………………………………………...[1]

(b) The amino acid shown below is isoleucine, 2-amino-3-methylpentanoic acid.



2-amino-3-methylpentanoic acid

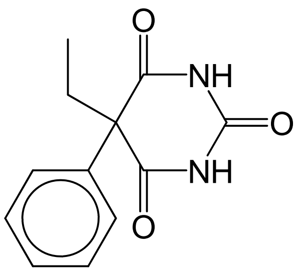
Molecule **F** can be used as the starting material to prepare this amino acid using a Strecker synthesis.

Draw the skeletal structure of **F**.

[1]

(c) An amide bond is formed when two amino acids react together.

Phenobarbital, which is a medication used to treat epilepsy, also has an amide bond in it.



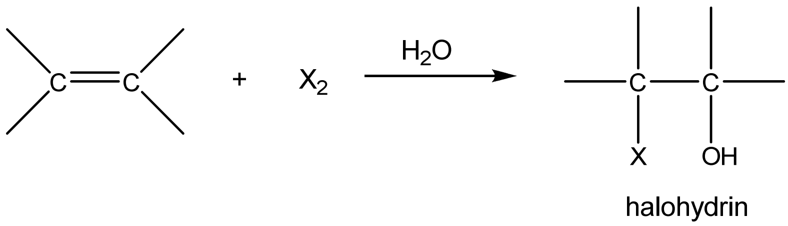
phenobarbital

Predict all the products formed when phenorbarbital undergoes acidic hydrolysis.

[2]

[Total: 6]

4 **(a)** Formation of 1,2-halo alcohols, also known as halohydrins, occurs via the addition reaction between an alkene and a halogen in the presence of water.



where X = C*l* or Br

In a series of experiments, the reaction between propene and aqueous bromine was carried out with different concentrations of the two reagents, and the following relative initial rates were obtained.

Experiment	$[CH_3CH{=}CH_2]$ / mol dm^{-3}	$[Br_2]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.020	0.020	1.00×10^{-3}
2	0.030	0.020	1.50×10^{-3}
3	0.040	0.030	3.00×10^{-3}

(i) Use these data to deduce the order of reaction with respect to each of the two reagents, showing how you arrive at your answers.

[2]

(ii) Hence write a rate equation for the reaction.

……………………………………………………………………………..…………….[1]

(iii) Calculate the rate constant for the reaction, giving its units.

[2]

(iv) State and explain how the rate of reaction may change if chlorine is used instead of bromine in the reaction with propene.

……………………………………………………………………………………………[1]

(b) **(i)** The mechanism of the addition reaction between propene and aqueous Br_2 involves three steps.

- There is an initial attack by the π electron pair of the alkene on Br_2 to yield a carbocation intermediate.
- This is followed by the nucleophilic attack of the lone pair of electrons on oxygen in water on the carbocation intermediate.
- The third step involves the loss of H^+ ion which then yields the neutral bromohydrin.

Using the information given above, describe a mechanism for this reaction.

[3]

(ii) Based on your mechanism drawn, explain whether it is consistent with the rate equation proposed.

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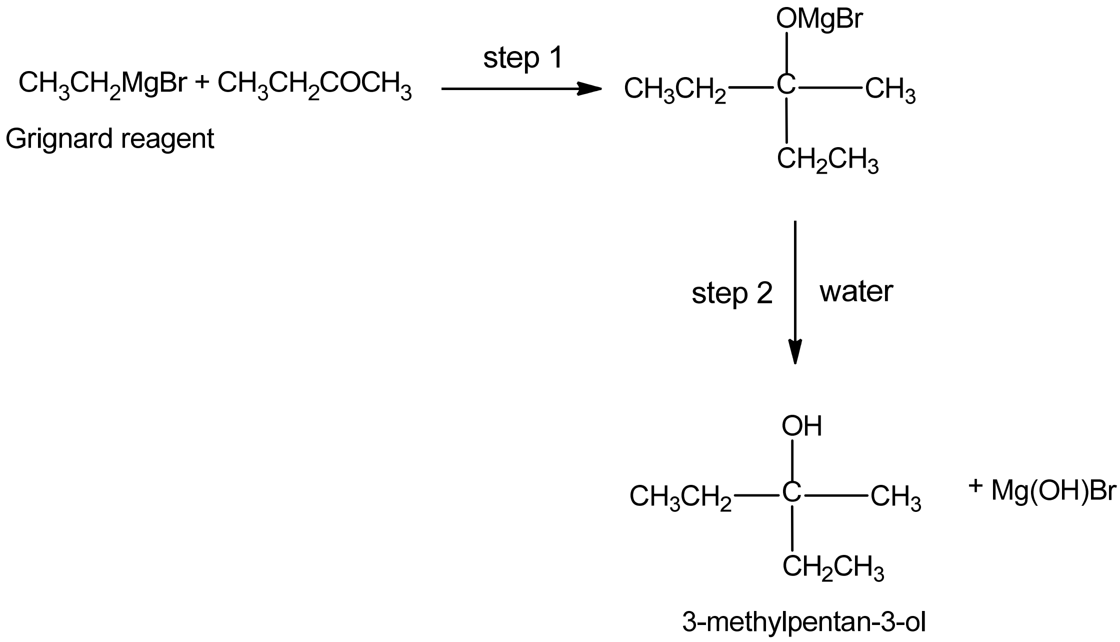
……………………………………………………………………………………………[1]

(c) Grignard reagents are organo-magnesium halides, commonly used in synthesis to prepare a variety of organic compounds.

The carbon-magnesium bonds in Grignard reagents are highly polar and this makes it extremely useful in organic synthesis as it is able to react with other polar organic molecules to form carbon-carbon bonds. An example of the use of a Grignard reagent is the two-step reaction of CH_3CH_2MgBr with butanone, $CH_3CH_2COCH_3$, to form 3-methylpentan-3-ol.



(i) Suggest the type of reaction that has taken place in step 1.

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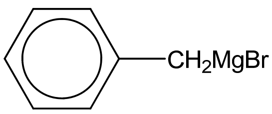
(ii) Suggest the structural formula of the final organic product formed when



is reacted with propanone, CH_3COCH_3, in a similar two-step process.

[1]

(iii) Suggest a suitable Grignard reagent and another organic compound to be used if propan-2-ol is to be prepared using a similar two-step process.

[2]

(iv) The Grignard reagent $CH_3CH_2CH_2MgBr$ can be readily converted into a carboxylic acid by using carbon dioxide.

Suggest the structural formula for the organic product formed.

[1]

[Total: 15]

5 **(a)** Cobalt forms many coloured complexes. Cobalt(III) chloride combines with ammonia to form a pink coloured compound **A**, $CoCl_3.H_2O.5NH_3$ (M_r = 268.4) in which the coordination number of cobalt is 6.

1.00 g of **A** is dissolved in 25 cm^3 of water and the solution is titrated with 0.500 mol dm^{-3} silver nitrate solution. It is found that 22.40 cm^3 of silver nitrate is required for complete reaction.

(i) Calculate the number of moles of free chloride ions per mole of **A**.

[2]

(ii) Draw the structure of the complex ion in **A.**

[1]

(iii) When the pink compound **A** is heated, water vapour and ammonia were evolved to give a purple solid **B**.
A and **B** have the same coordination number.

Suggest the formula of the complex in the purple solid **B**.

....[1]

(iv) Account for the difference in the colour of **A** and **B**.

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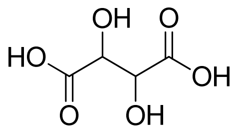
....[2]

(b) Aqueous hydrogen peroxide is fairly stable, but when a mixture of a cobalt(II) salt and tartaric acid is added to aqueous hydrogen peroxide, the initially pink solution slowly turns into a green cobalt(III) species.



tartaric acid

After a while, oxygen gas is vigorously evolved and the solution turns back to pink again.

(i) State the role of the cobalt(II) salt in this reaction and support your answer by referring to the observations.

Role of the cobalt(II) salt: ……………………………..…………………………….[1]

Observation with explanation:

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………………………………………..…………………………………………...[1]

(ii) Tartaric acid acts as a complexing agent in this reaction to stabilise the Co^{3+} cation.

With the aid of relevant data from the *Data Booklet*, show that Co^{3+} is not stable in aqueous solution.

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(c) A student wanted to measure the standard cell potential, $E^{\varnothing}_{cell}$, between the Co^{2+}/Co half-cell and the Fe^{3+}/Fe^{2+} half-cell. She set up and connected the two half cells and obtained a reading.

(i) Calculate the value $E^{\varnothing}_{cell}$ that will be obtained by the student.

[1]

(ii) State and explain what happens to the standard cell potential, $E^{\varnothing}_{cell}$ when ammonia is added to the Co^{2+}/Co half-cell.

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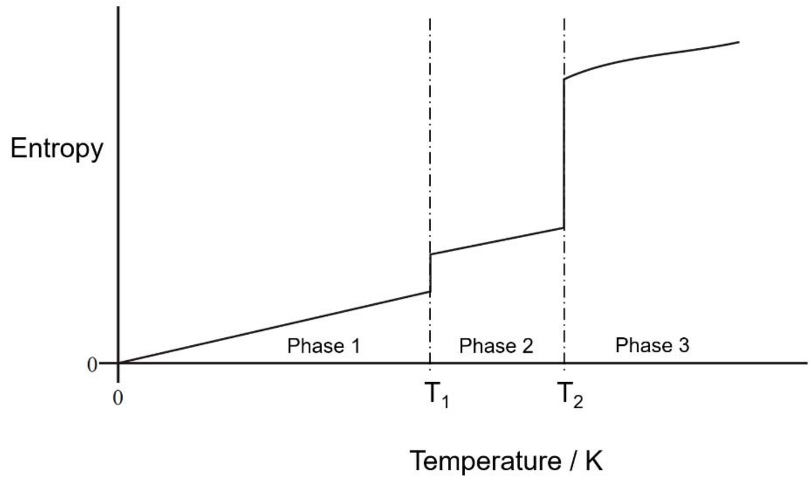
[Total: 13]

6 A student plotted the sketch graph below to show how the entropy of a sample of H_2O varies with temperature.



(a) Identify the state of H_2O in Phase 2 and 3 respectively. Suggest a value of T_2.

In phase 2:……………… In phase 3: ……………… value of T_2: ……………… [1]

(b) Suggest why entropy of H_2O is zero at 0 K.

……………………………………………………………………………………………[1]

(c) Explain why the entropy change, ΔS, at temperature T_2 is much larger than that at temperature T_1.

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……………………………………………………………………………………………[2]

(d) It requires 3.49 kJ of heat energy to convert 1.53 g of H_2O from the state in phase 2 to phase 3 at temperature T_2 and 100kPa.

Use these data and your value of T_2 in part **(a)** to calculate the value of ΔS, including units, for the conversion of one mole of H_2O from the state in phase 2 to phase 3 at temperature T_2.

[3]

(e) The student wants to find out if dissolving a salt, silver fluoride in water is always a spontaneous process. He must first find the enthalpy change of solution of silver fluoride in water.

Some enthalpy changes for silver fluoride are shown in the table.

	ΔH / kJ mol^{-1}
Lattice energy of silver fluoride	–950
Enthalpy change of hydration for silver ions	–464
Enthalpy change of hydration for fluoride ions	–506

(i) Use the data provided to calculate a value for the enthalpy change of solution of silver fluoride in water.

[2]

(ii) If entropy change for dissolving silver fluoride in water has a positive value, explain why dissolving of silver fluoride in water is always a spontaneous process.

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…………………………………………………………………………………………[2]

(iii) Explain why the enthalpy change of hydration of the fluoride ions is more negative than the enthalpy change of hydration of the chloride ions.

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…………………………………………………………………………………………[1]

[Total: 12]

7 The uses of carboxylic acids are so extensive that they can be divided into several industries, such as pharmaceuticals or food among others.

(a) Ethanol and ethanoic acid react reversibly to form ethyl ethanoate and water according to the equation:

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

A student mixed 0.0800 mol of ethanoic acid and 0.120 mol of ethanol in a conical flask and the flask was sealed with a bung and allowed to reach equilibrium at 20 °C.

The equilibrium mixture is placed in a graduated flask and the volume made up to 250 cm^3 with distilled water. A 10.0 cm^3 sample of this equilibrium mixture was placed in a conical flask with a few drops of phenolphthalein and titrated with 0.100 mol dm^{-3} of sodium hydroxide from a burette. The indicator turned pink when 6.40 cm^3 of NaOH had been added.

(i) Calculate the amount of CH_3COOH in the 250 cm^3 equilibrium mixture.

[1]

(ii) Hence, calculate the value for K_c for the reaction of ethanoic acid and ethanol at 20 °C.

[2]

(b) The following table compares the pK_a values of two dicarboxylic acids with that of ethanoic acid.

Acid	Formula	pK_1	pK_2
ethanoic	CH_3COOH	4.8	–
malonic	$HOOCCH_2COOH$	2.8	5.7
succinic	$HOOC(CH_2)_2COOH$	4.2	5.6

(i) Suggest a reason why the pK_1 value of malonic acid is so much less than the pK_1 of ethanoic and succinic acid.

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……………………………………………………………………………………...…[2]

(ii) Suggest a reason why the pK_2 value of malonic and succinic acid is higher than its respective pK_1 value.

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……………………………………………………………………………………...…[1]

(c) Malonic acid can undergo dehydration with P_4O_{10} to give a foul-smelling gas, **A**. At 30.5 kPa, 0.1057 g of A occupies 200 cm^3 at a temperature of 200^{o}C. Determine the relative molecular mass of **A**.

[2]

(d) At high temperature, succinic acid can also undergo dehydration to produce a neutral compound **B**, $C_4H_4O_3$ which does not react with sodium metal or Brady's reagent.

Compound **B** reacts with ammonia to give a compound **C**, $C_4H_7NO_3$, which reacts with cold NaOH(aq), but not with cold HC*l*(aq).

Suggest structures for **B** and **C** and explain the observations.

[4]

[Total: 12]

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